

A LOCAL LEMMA REFINEMENT OF THE BALOGH–EBERHARD–NARAYANAN–TREGLOWN–WAGNER LOWER BOUND FOR FOLKMAN’S THEOREM

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ABSTRACT. Folkman’s theorem asserts that for each $k \in \mathbb{N}$, there exists a natural number $n = F(k)$ such that whenever the elements of $[n]$ are two-coloured, there exists a set $A \subset [n]$ of size k with the property that all the sums of the form $\sum_{x \in B} x$, where B is a nonempty subset of A , are contained in $[n]$ and have the same colour. Balogh, Eberhard, Narayanan, Treglown and Wagner [2] proved that $F(k) \geq 2^{2^{k-1}/k}$ for all $k \in \mathbb{N}$. Their proof uses a random colouring based on the 2-adic valuation, followed by a union bound over all k -sets. We observe that the same random colouring combined with the Lovász local lemma gives the stronger estimate

$$F(k) \geq 2^{2^{k-1}/(k-1)-O(1)}.$$

Equivalently, for some absolute constant $C > 0$ and all sufficiently large k ,

$$F(k) > 2^{(2^{k-1}-Ck)/(k-1)}.$$

1. INTRODUCTION

Schur’s theorem, proved in 1916, is one of the central results of Ramsey theory and asserts that whenever the elements of $\mathbb{N}$ are finitely coloured, there exists a monochromatic set of the form $\{x, y, x + y\}$ for some $x, y \in \mathbb{N}$. About fifty years ago, a wide generalisation of Schur’s theorem was obtained independently by Folkman, Rado and Sanders, and this generalisation is now commonly referred to as Folkman’s theorem (see [4], for example). To state Folkman’s theorem, it will be convenient to have some notation. For $n \in \mathbb{N}$, we write $[n]$ for the set $\{1, 2, \dots, n\}$, and for a finite set $A \subset \mathbb{N}$, let

$$S(A) = \left\{ \sum_{x \in B} x : B \subset A \text{ and } B \neq \emptyset \right\}$$

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denote the set of all *finite sums* of A . In this language, Folkman's theorem states that for all $k, r \in \mathbb{N}$, there exists a natural number $n = F(k, r)$ such that whenever the elements of $[n]$ are r -coloured, there exists a set $A \subset [n]$ of size k such that $S(A)$ is a monochromatic subset of $[n]$; of course, it is easy to see that Folkman's theorem, in the case where $k = 2$, implies Schur's theorem.

In this note, we shall be concerned with lower bounds for the two-colour Folkman numbers, i.e., for the quantity $F(k) = F(k, 2)$. All logarithms are to the base 2. In 1989, Erdős and Spencer [3] proved that

$$F(k) \geq 2^{ck^2/\log k} \quad (1)$$

for all $k \in \mathbb{N}$, where $c > 0$ is an absolute constant. This bound stood until the recent breakthrough of Balogh, Eberhard, Narayanan, Treglown and Wagner [2], who established the doubly exponential lower bound

$$F(k) \geq 2^{2^{k-1}/k} \quad (2)$$

for all $k \in \mathbb{N}$. Their proof introduces a clever random colouring based on the 2-adic valuation in which colours alternate along geometric progressions of the form $q, 2q, 4q, \dots$ for odd q . They show that any k -set A with $S(A)$ monochromatic must have all $2^k - 1$ subset sums distinct, and that the monochromatic event then has probability at most $2^{1-2^{k-1}}$. A union bound over at most $(en/k)^k$ candidate sets completes their argument.

Our contribution is the observation that the union bound in the argument of [2] can be replaced by the Lovász local lemma. The reason this is possible is that the event $\{S(A) \text{ is monochromatic}\}$ depends only on the random bits indexed by the odd parts of the elements of $S(A)$, and a fixed odd integer can appear as the odd part of a subset sum for only polynomially many candidate sets. This yields an improved exponent.

Theorem 1.1. *There is an absolute constant $C > 0$ such that, for all sufficiently large k ,*

$$F(k) > 2^{(2^{k-1}-Ck)/(k-1)}. \quad (3)$$

Consequently,

$$\log F(k) \geq \frac{2^{k-1}}{k-1} - O(1). \quad (4)$$

The gain in the exponent over (2) is

$$\frac{2^{k-1}}{k-1} - \frac{2^{k-1}}{k} - O(1) = \frac{2^{k-1}}{k(k-1)} - O(1),$$

so the lower bound improves by a multiplicative factor of $2^{2^{k-1}/(k(k-1)) - O(1)}$.

This short note is organised as follows. We recall the random colouring of [2] and its key probabilistic estimate in Section 2, give the proof of Theorem 1.1 in Section 3, and conclude with some remarks in Section 4.

2. THE BALOGH–EBERHARD–NARAYANAN–TREGLOWN–WAGNER COLOURING

We recall the random colouring introduced in [2]. Independently colour each odd integer $q \leq N$ with a random bit $X_q \in \{0, 1\}$. For each $n \in [N]$, write $n = 2^{v_2(n)} \text{odd}(n)$, where $\text{odd}(n)$ is odd, and define

$$\chi(n) = X_{\text{odd}(n)} + v_2(n) \pmod{2}.$$

Thus the colours alternate along every geometric progression $q, 2q, 4q, 8q, \dots$ with q odd.

We need the following estimate, which is the central probabilistic ingredient of [2].

Claim 2.1. *For every k -set $A \subset [N]$ with $S(A) \subset [N]$,*

$$\mathbb{P}(S(A) \text{ is monochromatic}) \leq 2^{1-2^{k-1}}.$$

Proof. We include the short argument for completeness.

If $|S(A)| \leq 2^k - 2$, then two distinct nonempty subsets $B_1, B_2 \subseteq A$ have the same sum. Removing $B_1 \cap B_2$ from both, we may assume B_1 and B_2 are disjoint and still have equal positive sums. Let $x = \sum_{a \in B_1} a = \sum_{a \in B_2} a$. Then $x \in S(A)$ and $2x = \sum_{a \in B_1 \cup B_2} a \in S(A)$, but the colouring satisfies $\chi(2x) \neq \chi(x)$. Hence $S(A)$ cannot be monochromatic, and the probability is 0.

Now suppose $|S(A)| = 2^k - 1$, so all nonempty subset sums are distinct. For each odd $m \in \mathbb{N}$, define $G_m = \{m, 2m, 4m, \dots\} \cap [N]$; these progressions partition $[N]$. Let $s = \min_{a \in A} v_2(a)$ and write $a = 2^s a'$ for each $a \in A$. At least one of the integers a' is odd. Among the 2^k subsets of A , exactly half yield

$$\sum_{a \in B} a' \equiv 1 \pmod{2},$$

and the empty subset is not among them. Hence there are exactly 2^{k-1} nonempty subsets $B \subseteq A$ for which $v_2(\sum_{a \in B} a) = s$. The corresponding 2^{k-1} sums are distinct and, having the same 2-adic valuation, lie in distinct progressions G_m .

Define $B_A \subseteq S(A)$ by choosing exactly one element from each progression G_m that meets $S(A)$. Then $|B_A| \geq 2^{k-1}$. The colours of the elements of B_A are independent fair bits, since they depend on distinct random variables X_m . Therefore

$$\mathbb{P}(B_A \text{ is monochromatic}) = 2^{1-|B_A|} \leq 2^{1-2^{k-1}}.$$

Since $S(A)$ monochromatic implies B_A monochromatic, the result follows. $\square$

3. PROOF OF THE MAIN RESULT

In this section, we give the proof of Theorem 1.1. The key tool is the symmetric form of the Lovász local lemma (see [1], for example).

Lemma 3.1 (Symmetric Lovász local lemma). *Let $\{E_i\}_{i \in I}$ be a finite family of events. Suppose each event has probability at most p , and each event is mutually independent of all but at most D other events. If $ep(D+1) \leq 1$, then $\mathbb{P}(\bigcap_{i \in I} E_i^c) > 0$.*

Proof of Theorem 1.1. Let N be an integer to be chosen later, and use the random colouring described in Section 2. Define

$$\mathcal{A} = \left\{ A \subset \mathbb{N} : |A| = k, \sum_{a \in A} a \leq N \right\}.$$

It suffices to consider this family, since $S(A) \subset [N]$ implies $\sum_{a \in A} a \leq N$. For each $A \in \mathcal{A}$, define the bad event

$$E_A = \{S(A) \text{ is monochromatic}\}.$$

By Claim 2.1,

$$\mathbb{P}(E_A) \leq p := 2^{1-2^{k-1}}. \tag{5}$$

For $A \in \mathcal{A}$, define $O(A) = \{\text{odd}(s) : s \in S(A)\}$. The event E_A depends only on the random variables $\{X_q : q \in O(A)\}$. Hence E_A is mutually independent of every collection of events $\{E_B : O(A) \cap O(B) = \emptyset\}$. We define a dependency graph on $\mathcal{A}$ by joining A and B whenever $O(A) \cap O(B) \neq \emptyset$, and now bound its maximum degree.

Fix $A \in \mathcal{A}$. Since $|S(A)| \leq 2^k - 1$, we have $|O(A)| \leq 2^k - 1$. Fix an odd integer $q \in O(A)$. We count the number of $B \in \mathcal{A}$ such that $q \in O(B)$.

The condition $q \in O(B)$ means that there exists a nonempty subset $T \subseteq B$ and an integer $j \geq 0$ such that $\sum_{t \in T} t = q2^j$. Put $M = q2^j$; since $B \in \mathcal{A}$, we have $M \leq N$. Suppose $|T| = r$. For fixed M and r , the number of possible sets T is at most $\binom{M-1}{r-1}$, and the remaining $k - r$ elements of B contribute total sum at most $N - M$, giving at most $\binom{N-M}{k-r}$ possibilities (this bound is also valid when $r = k$). Therefore

$$\#\{B \in \mathcal{A} : q \in O(B)\} \leq \sum_{r=1}^k \sum_{\substack{j \geq 0 \\ q2^j \leq N}} \binom{q2^j - 1}{r-1} \binom{N - q2^j}{k-r}. \quad (6)$$

We estimate this sum. For $r = 1$,

$$\sum_{\substack{j \geq 0 \\ q2^j \leq N}} \binom{N - q2^j}{k-1} \leq (\log N + 1)N^{k-1}.$$

For $r \geq 2$, we use $\binom{q2^j-1}{r-1} \leq (q2^j)^{r-1}$ and $\binom{N-q2^j}{k-r} \leq N^{k-r}$. The inner geometric sum satisfies

$$\sum_{\substack{j \geq 0 \\ q2^j \leq N}} (q2^j)^{r-1} \leq 2N^{r-1},$$

so the r -th contribution is at most $2N^{k-1}$. Summing over $r = 2, \dots, k$ gives at most $2kN^{k-1}$. Combining with the $r = 1$ term, for $k \geq 1$ and $N \geq 2$,

$$\#\{B \in \mathcal{A} : q \in O(B)\} \leq 3k(\log N + 1)N^{k-1}. \quad (7)$$

Since $|O(A)| \leq 2^k - 1$, the maximum degree in the dependency graph satisfies

$$D \leq 3k \cdot 2^k (\log N + 1)N^{k-1}. \quad (8)$$

Now choose

$$N = \left\lfloor 2^{(2^{k-1}-Ck)/(k-1)} \right\rfloor \quad (9)$$

where $C > 0$ is a sufficiently large absolute constant. Then $N^{k-1} \leq 2^{2^{k-1}-Ck}$ and $\log N + 1 \leq 2^k$ for all sufficiently large k . Hence

$$D \leq 3k \cdot 2^k \cdot 2^k \cdot 2^{2^{k-1}-Ck}.$$

Since $3k \leq 2^{2k}$ for all sufficiently large k , we obtain

$$D \leq 2^{2^{k-1}-(C-4)k}. \quad (10)$$

Recalling (5), for sufficiently large k ,

$$ep(D+1) \leq e \cdot 2^{1-2^{k-1}} \left(2^{2^{k-1}-(C-4)k} + 1 \right) \leq 2^{-(C-5)k}.$$

Choosing C sufficiently large, for example $C = 10$, gives $ep(D+1) < 1$. By Lemma 3.1, with positive probability none of the events E_A occurs. Therefore there exists a two-colouring of $[N]$ such that no k -set A has $S(A)$ monochromatic. Hence

$$F(k) > N = \left\lfloor 2^{(2^{k-1}-Ck)/(k-1)} \right\rfloor,$$

proving (3) and consequently (4). $\square$

4. CONCLUDING REMARKS

We conclude this note with a few remarks.

First, we emphasise that the argument uses exactly the same random colouring as Balogh–Eberhard–Narayanan–Treglown–Wagner [2]. The improvement comes entirely from replacing the global union bound over all k -sets by a local lemma argument. This is possible because the event $\{S(A) \text{ is monochromatic}\}$ depends only on the random bits indexed by the odd parts of the elements of $S(A)$, and a fixed odd part can occur in the subset sums of only $O_k((\log N)N^{k-1})$ relevant k -sets.

Second, while our bound (4) improves the exponent from $2^{k-1}/k$ to $2^{k-1}/(k-1) - O(1)$, a multiplicative gain of $2^{2^{k-1}/(k(k-1)) - O(1)}$, this lower bound remains considerably far from the best known upper bound for $F(k)$, which is of tower type; see [6], for instance. Closing this gap remains a major open problem.

Third, we note that the dependency bound (8) is unlikely to be tight, and a more careful analysis of the dependency structure may yield further modest improvements. However, any approach based on the colouring of [2] is fundamentally limited by the probability estimate in Claim 2.1, and substantially improving the lower bound for $F(k)$ would likely require new colouring constructions.

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